

Data Scientist and founder of LúthTech Performance, with a PhD in Data Science specialising in applied machine learning and analytics for athlete monitoring, performance evaluation, and workload management in elite sport. Experienced in designing and delivering end-to-end data science solutions, from raw sensor and tracking data ingestion through feature engineering, modelling, validation, and deployment into production analytics workflows. Postdoctoral work focused on building scalable, cloud-based (Azure) data platforms integrating GNSS/IMU, wellness, and clinical data sources (e.g. STATSports, Kitman, VALD) to generate reliable, actionable performance insights. Currently applying this expertise through LúthTech Performance, an independent practice supporting sporting organisations with athlete monitoring, performance analytics, and data management systems.

SKILLS

PhD in Data Science , <i>Centre for Research and Training in Artificial Intelligence (CRT-AI), Dublin City University</i> 2020 — 2025	Python, R, SQL, Git, $\LaTeX$, Markdown
Programming & Tools	
Machine Learning	Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch (model development, validation, deployment)
Data Engineering & ETL	Pandas, Polars, NumPy, SQL-based pipelines, Spark, data quality validation, feature engineering
Databases & Cloud	MySQL, SQL Server, NoSQL concepts, Azure-based data platforms
Data Visualisation	Matplotlib, Seaborn, Plotly, Power BI, Tableau
Applied Analytics & Communication	Translating analytical outputs into decision-ready insights for technical and non-technical stakeholders

TECHNICAL EXPERIENCE

Founder & Data Scientist <i>LúthTech Performance</i>	Sep 2026 — Present <i>Dublin, Ireland</i>
- Applying machine learning and AI to athlete monitoring, building models that use GNSS and wearable sensor data to improve performance and analytics in team sport - Bringing hands-on experience in digital transformation and data management systems to sporting organisations, drawing on postdoctoral platform-delivery work - Continuing applied research into performance systems design	
Career Break — Full-time Parenting	Jul 2026 — Present <i>Ireland</i>
Took a planned career break to be present for the arrival of my first child, while laying the groundwork for LúthTech Performance	
Data Scientist <i>STATSports</i>	May 2026 — Jul 2026 <i>Newry, Northern Ireland</i>
Developed machine learning models for athlete monitoring in sport-specific scenarios, applying ML techniques to wearable sensor-derived data	
Postdoctoral Researcher <i>Insight Centre for Data Analytics Sport Ireland Institute</i>	Feb 2025 — Apr 2026 <i>Dublin, Ireland</i>
- Leading the design and delivery of a production-grade Azure analytics platform to centralise multi-source athlete monitoring data (GNSS, IMU, VALD, Kitman Labs) across national squads - Designed scalable data-lake architectures (raw, staging, curated layers) and automated ETL pipelines using Python and Azure Data Factory - Built Synapse Analytics SQL pools and serverless views to power near real-time Power BI dashboards used daily by coaches, sports scientists, and medical staff - Implemented reproducible data engineering and analytics standards, including version-controlled pipelines, data-quality validation, and governance frameworks - Partnered closely with high-performance stakeholders to translate complex sensor and workload data into actionable insights for fatigue management, injury-risk reduction, and return-to-performance decisions	
PhD Researcher in Data Science <i>Centre for Research and Training in AI, Dublin City University & KU Leuven</i>	Oct 2020 — Mar 2025 <i>Ireland & Belgium</i>
- Built end-to-end data pipelines processing 100+ GB of high-frequency GNSS and IMU time-series data using Python (Pandas, Dask) and scalable computing workflows - Developed and validated predictive machine-learning models (XGBoost, Random Forest, Gradient Boosting) achieving 85%+ accuracy for workload and performance-related outcomes	

- Designed advanced time-series feature engineering pipelines, reducing feature dimensionality by 60% through multicollinearity analysis and recursive feature elimination while maintaining model performance
- Collaborated with international teams to develop hybrid modelling approaches combining engineered features and deep learning for longitudinal performance prediction
- Built interactive dashboards and automated reports (Tableau, Plotly) to communicate analytical outputs to non-technical performance stakeholders
- Published peer-reviewed research and presented at international conferences, bridging applied machine learning with real-world elite sports performance challenges

Sport Scientist

Oct 2018 — Jun 2025

Cavan GAA & Offaly Senior Football

Ireland

- Designed and deployed an end-to-end athlete performance monitoring system integrating GNSS data, subjective wellness inputs, and automated analytics pipelines for 50+ athletes
- Processed and analysed large-scale longitudinal training-load datasets using Python (Pandas, NumPy) to extract performance and fatigue indicators
- Automated weekly reporting and dashboard workflows, reducing reporting turnaround from hours to minutes
- Implemented efficient processing strategies for high-frequency sensor data (10Hz to 2Hz) to support scalable season-long analysis
- Worked directly with coaches, S&C staff, and medical teams to translate analytical outputs into actionable training and recovery interventions
- Applied statistical analysis and predictive modelling to evaluate training effectiveness and optimise workload distribution across competitive seasons

Selected Focus Areas

- Athlete monitoring, workload modelling, and performance evaluation
- GNSS / IMU sensor data analytics and time-series feature engineering
- Predictive modelling for performance optimisation and injury-risk mitigation
- Cloud-based data platforms and production-ready analytics pipelines
- Translating complex analytics into actionable insights for coaches and medical staff

EDUCATION

PhD in Data Science, *Centre for Research and Training in Artificial Intelligence (CRT-AI), Dublin City University* 2020 — 2024

- Thesis: Applied machine learning for athlete monitoring, focusing on prediction of Rating of Perceived Exertion and oxygen uptake from GNSS / IMU-derived features in team sports

MSc by Research in Exercise Physiology, *Dublin City University* 2018 — 2020

- Thesis: [Effects of match play on neuromuscular performance, muscle damage, and fatigue markers in elite Gaelic football](#)

Bachelor of Science in Physical Education (Biology) & Higher Diploma in Education, *Dublin City University* 2006 — 2010

ACTIVITIES

Selected Activities & Contributions

ECML PKDD – MLSA24 Workshop, Presenter on automated shot-event detection from GNSS sensor data, demonstrating applied machine-learning pipelines for high-frequency wearable tracking data (Vilnius, Lithuania) [\[View Online\]](#) Sep 2024

PLOS ONE (Peer-Reviewed Journal), Author of applied machine-learning study estimating oxygen uptake in simulated team sports using wearable sensor data (research completed as Visiting Scholar, KU Leuven) [\[View Online\]](#) Feb 2025

Biology of Sport (Peer-Reviewed Journal), Author of applied machine-learning study on predicting Ratings of Perceived Exertion in elite team-sport athletes using tracking-derived features [\[View Online\]](#) Jan 2024

ACSM World Congress, Poster presentation on translating match-load data into physiological and neuromuscular performance insights; published abstract in *Medicine and Science in Sports and Exercise* Jun 2017